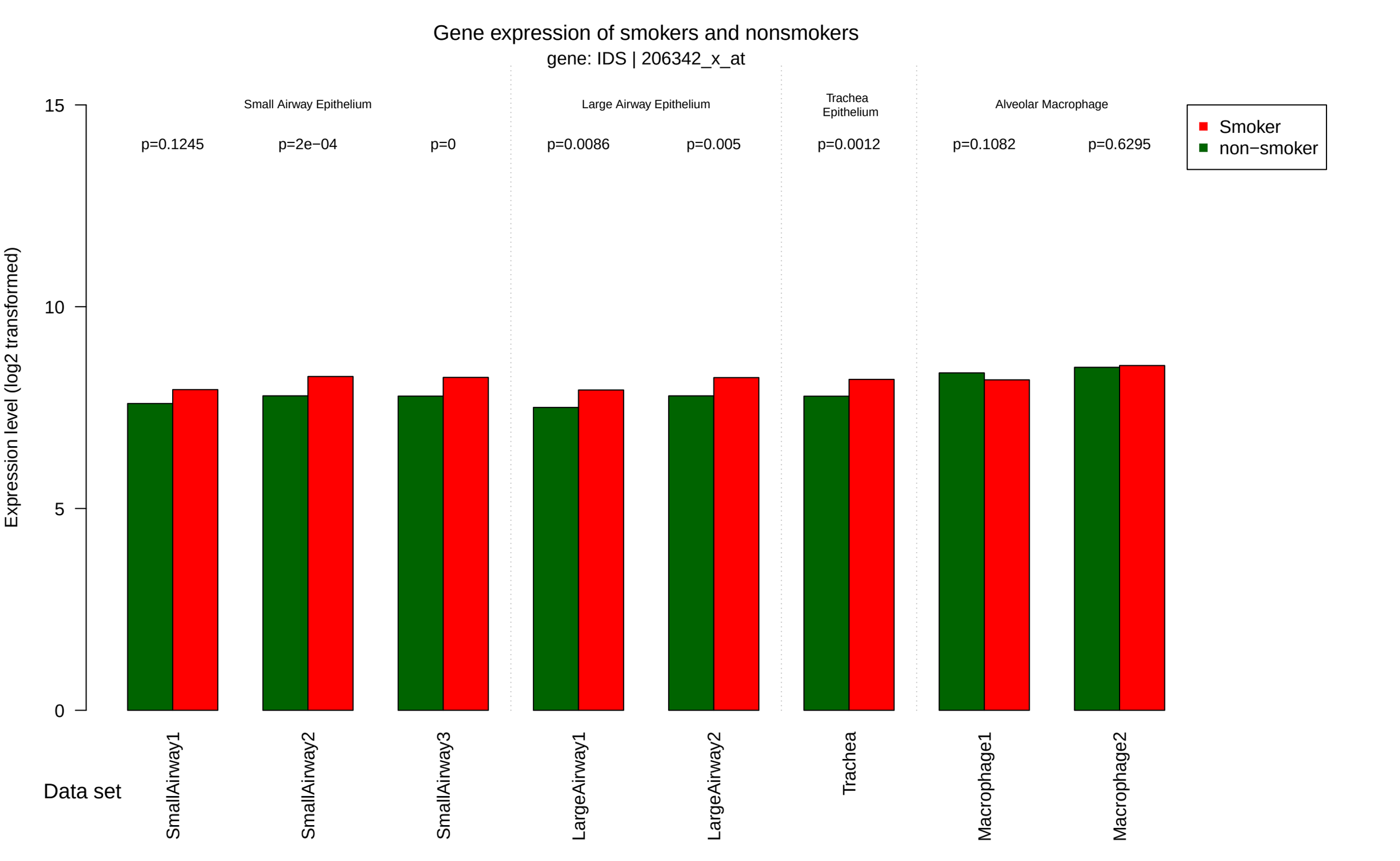
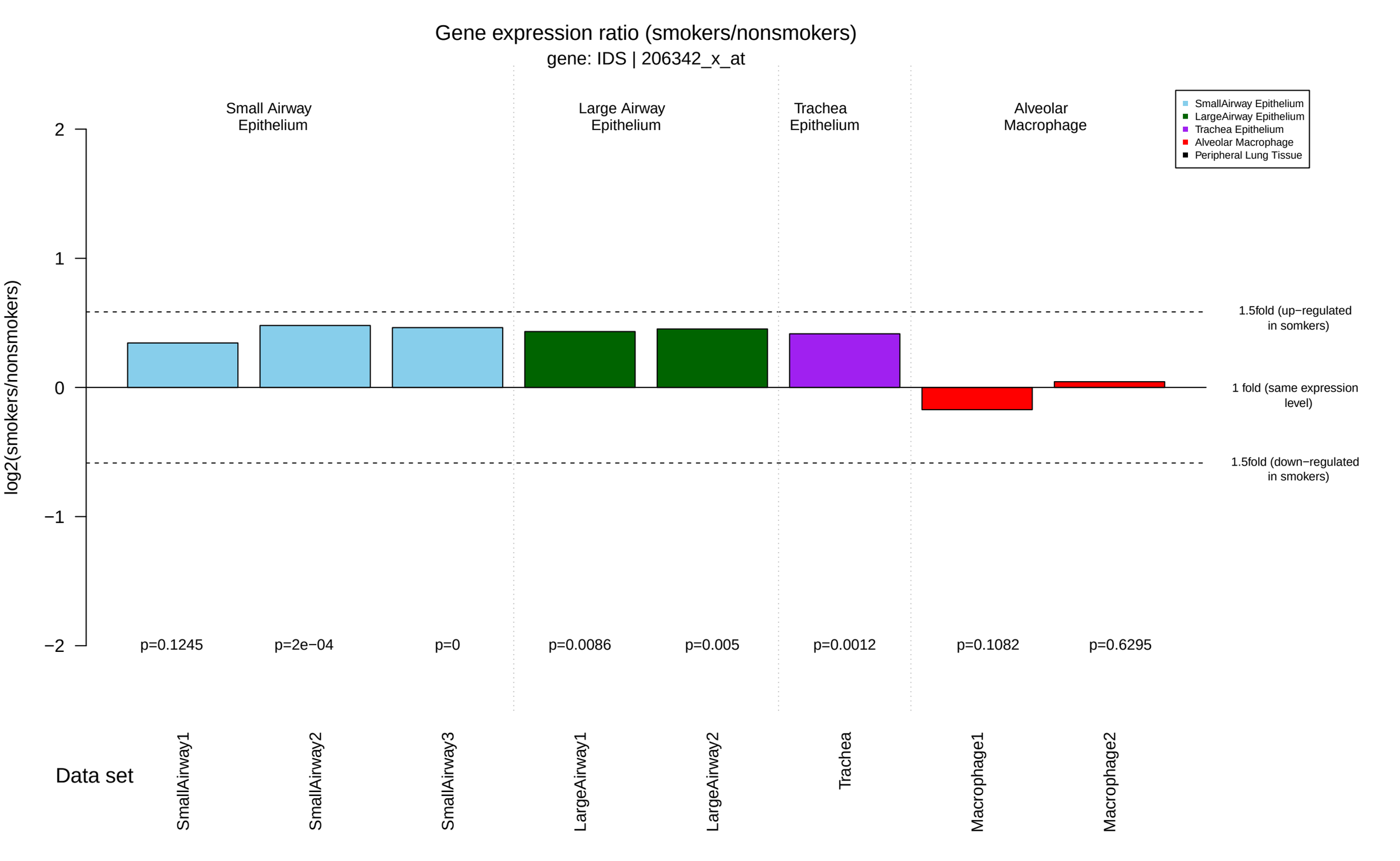
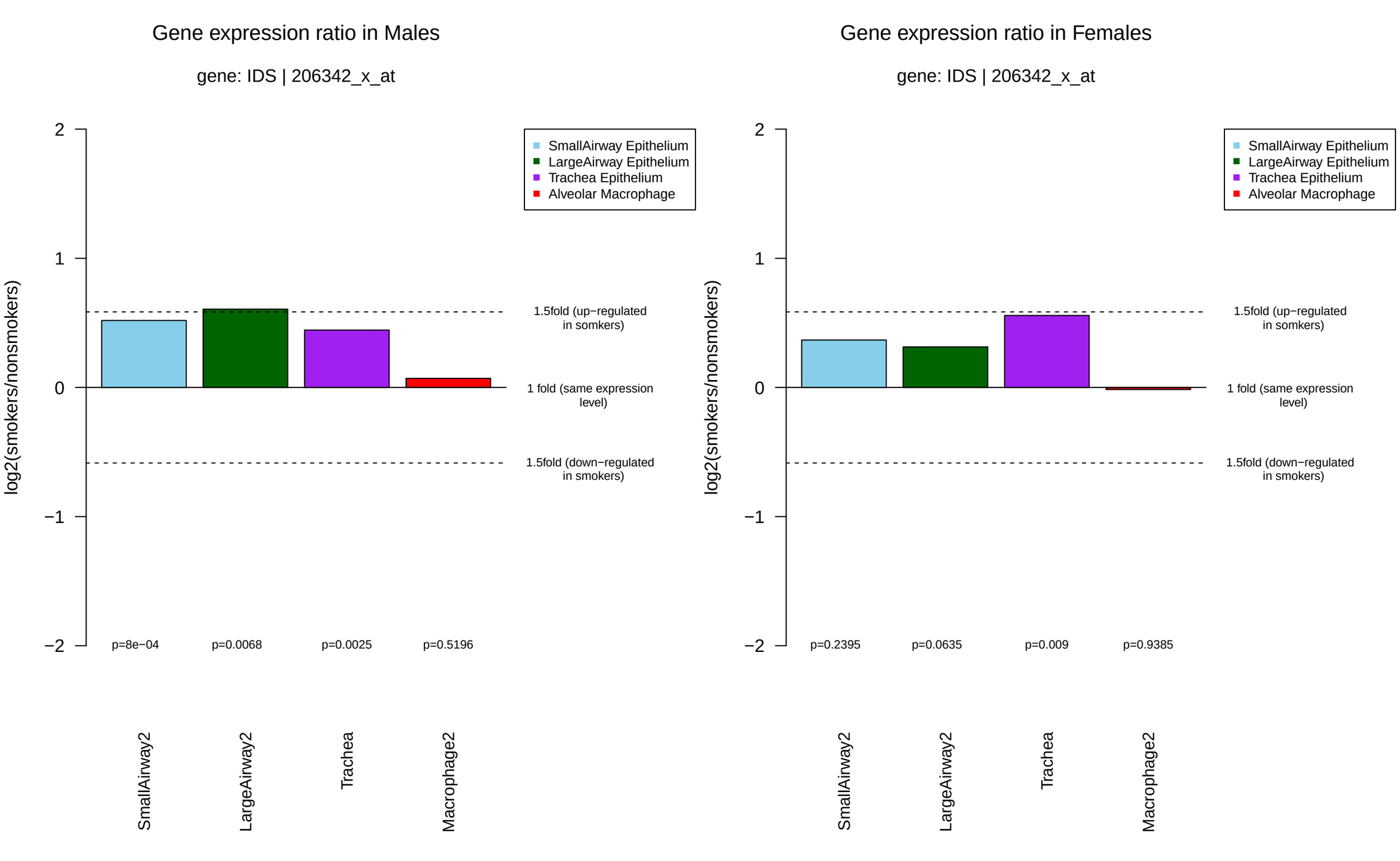
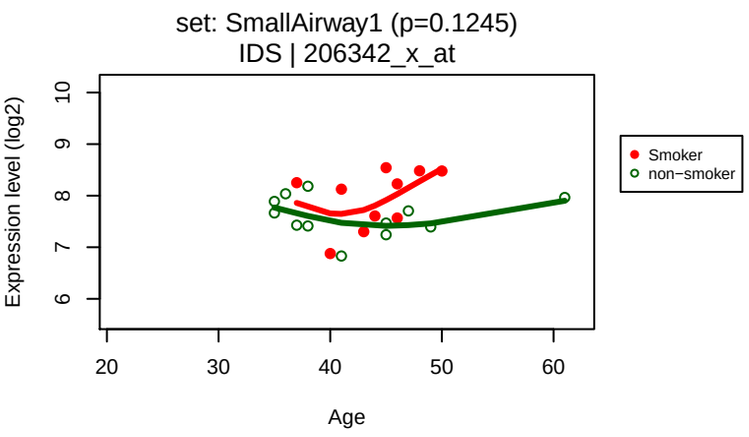
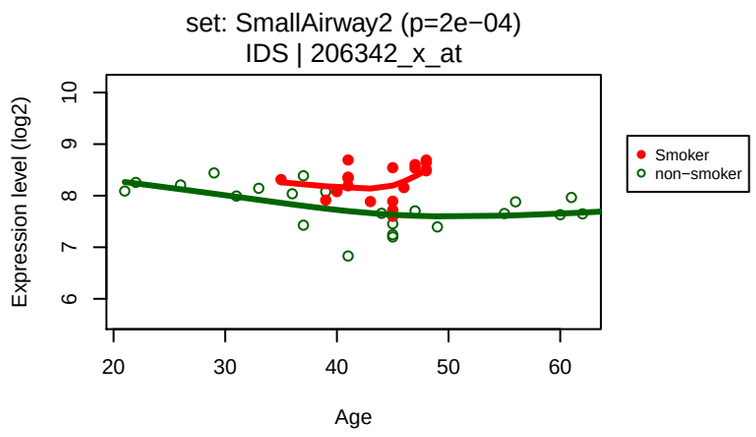
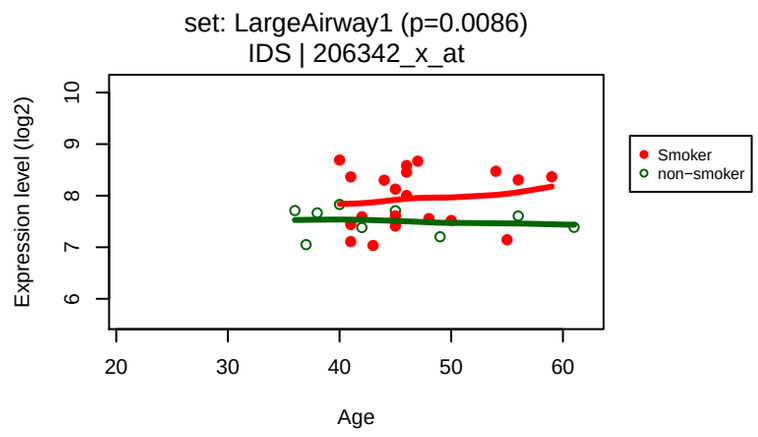

Gene expression of smokers and nonsmokers
gene: IDS | 206342_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1245
p=2e−04
p=0
p=0.0086
p=0.005
p=0.0012
p=0.1082
p=0.6295
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

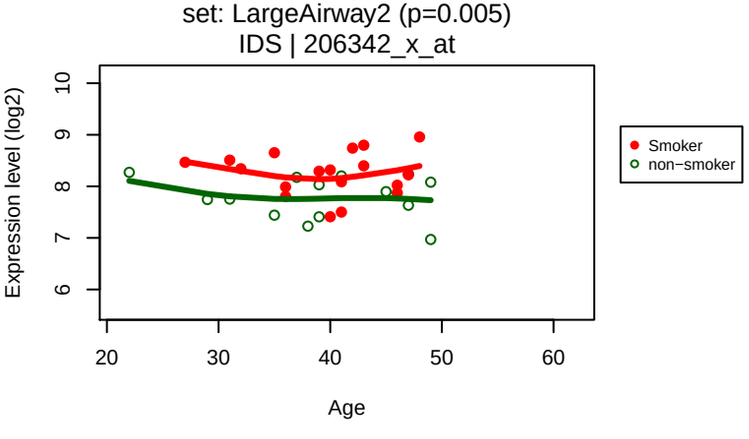

Gene expression ratio (smokers/nonsmokers)
gene: IDS | 206342_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1245
p=2e−04
p=0
p=0.0086
p=0.005
p=0.0012
p=0.1082
p=0.6295
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

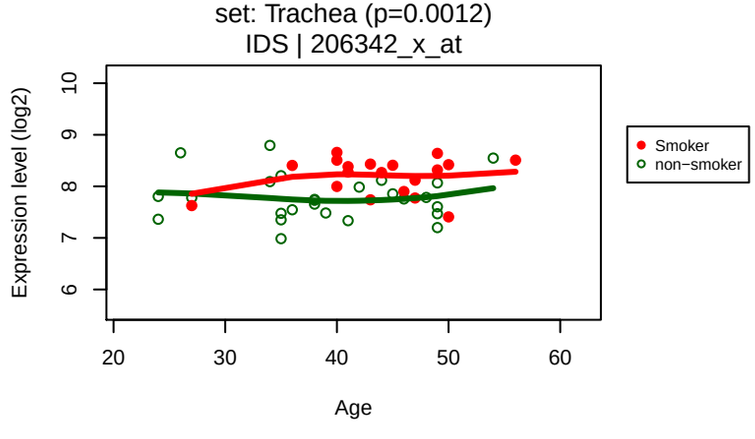

Gene expression ratio in Males
gene: IDS | 206342_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=8e−04
p=0.0068
p=0.0025
p=0.5196
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: IDS | 206342_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2395
p=0.0635
p=0.009
p=0.9385
SmallAirway2
LargeAirway2
Trachea
Macrophage2

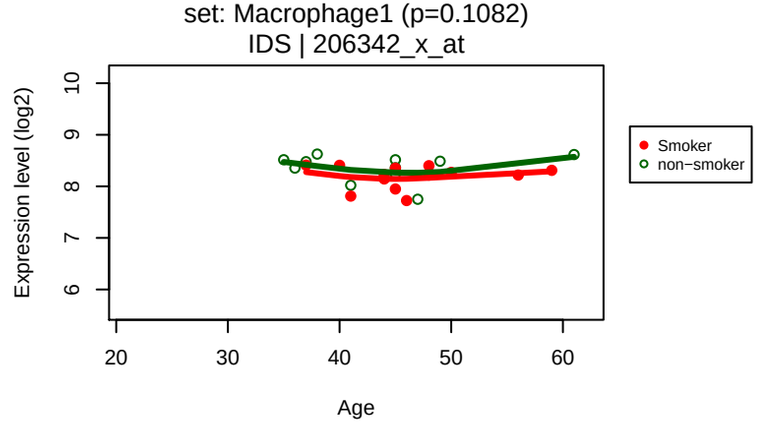

set: SmallAirway1 (p=0.1245)
IDS | 206342_x_at
Expression level (log2)
Age
Smoker
non−smoker

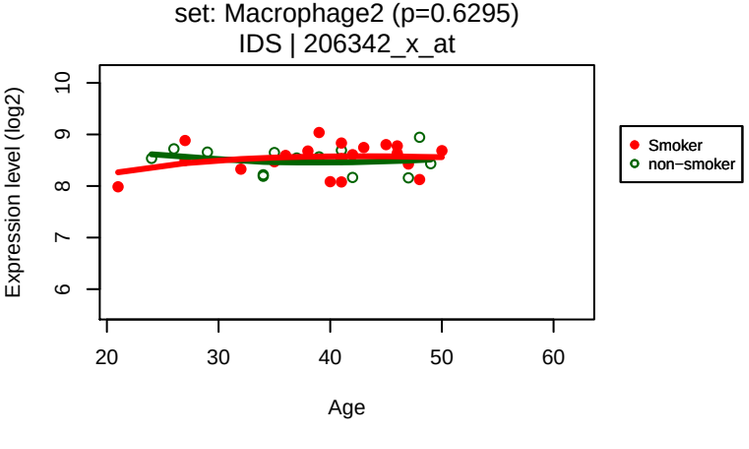

set: SmallAirway2 (p=2e−04)
IDS | 206342_x_at
Expression level (log2)
Age
Smoker
non−smoker

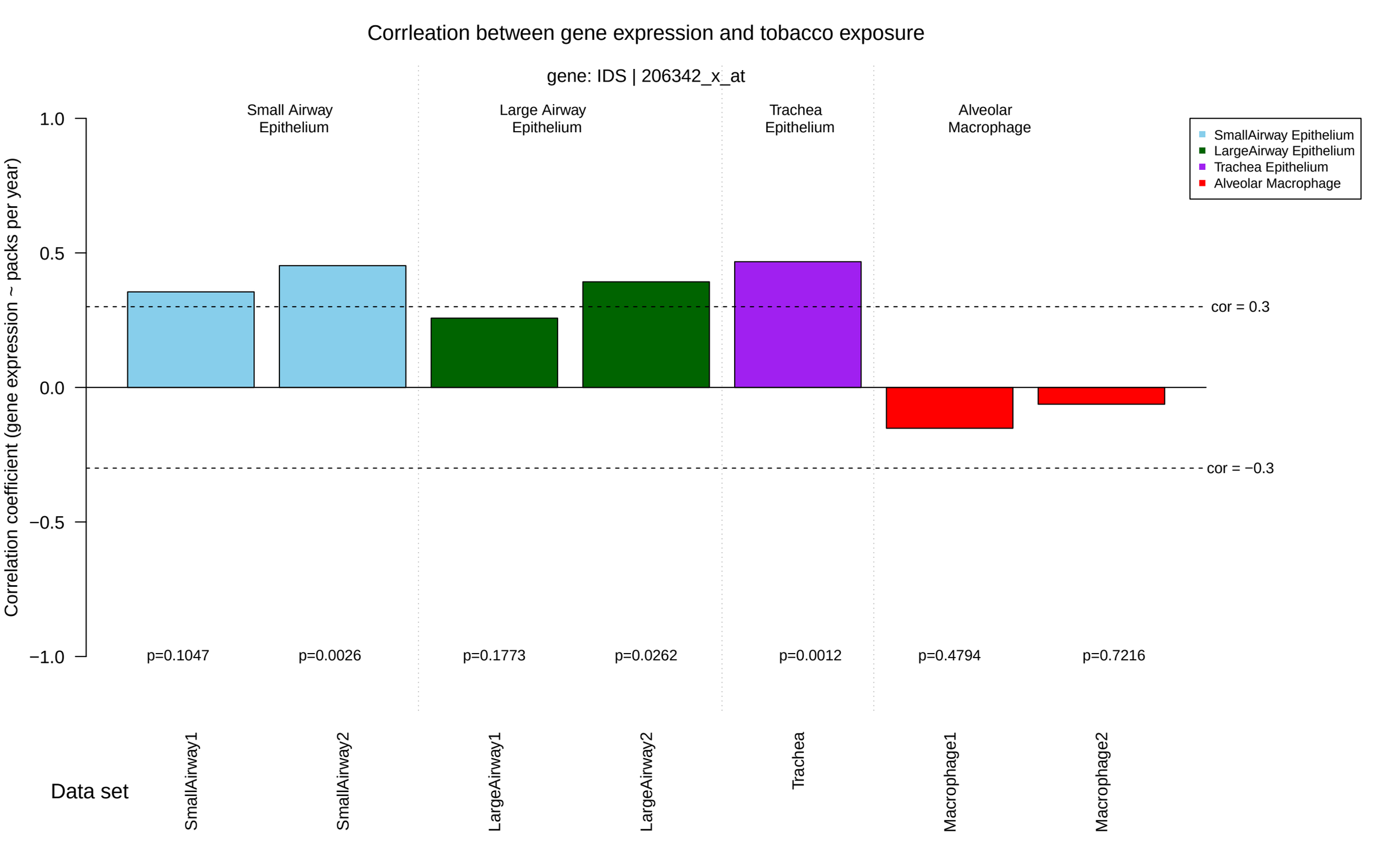

set: LargeAirway1 (p=0.0086)
IDS | 206342_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.005)
IDS | 206342_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0012)
IDS | 206342_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1082)
IDS | 206342_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6295)
IDS | 206342_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: IDS | 206342_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1047
p=0.0026
p=0.1773
p=0.0262
p=0.0012
p=0.4794
p=0.7216
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2